Week 2 Homework + Weekly Quiz

Sampling, Aliasing, Reconstruction, D/A, and Quantization

DSP + ML • Combined senior / first-year graduate ECE

Text alignment: Oppenheim & Schafer, Discrete-Time Signal Processing

Homework	8 core problems + 1 graduate stretch problem
Quiz	20 points; designed for about 15 minutes
Solution key	Complete reasoning, calculations, and grading guidance
Code connection	Problems reinforce the Week2_DSP_ML_Code notebook and audio demos

Homework — Student Copy

Assume ideal mathematical models unless a problem explicitly asks about a practical filter or quantizer. Show enough work to make your frequency mappings and assumptions unambiguous.

Problem 1. From continuous frequency to discrete frequency [10 points]

A microphone voltage is $x_c(t) = 2\cos(2\pi\cdot 1000\, t) + \cos(2\pi\cdot 5000\, t)$. It is sampled at $f_s = 16$ kHz.

1. (a) Write $x[n]$ explicitly.
2. (b) Give the two discrete-time angular frequencies in rad/sample and as multiples of π.
3. (c) Are either components aliased at this sample rate? Explain.

Problem 2. Spectral replicas and Nyquist margin [12 points]

An analog signal is strictly bandlimited to $|f| \le 3.4$ kHz and is sampled at $f_s = 8$ kHz.

4. (a) Sketch or describe the locations of the baseband spectrum and the first two pairs of replicas after ideal sampling.
5. (b) What is the separation between adjacent spectral replicas?
6. (c) What guard-band width exists between the positive edge of the baseband copy and the nearest edge of the $+f_s$ replica?
7. (d) What is the ideal Nyquist rate for this signal? Is 8 kHz above it?

Problem 3. Alias-frequency table [12 points]

Each row is a real sinusoid sampled at $f_s = 12$ kHz. For each analog frequency, find the observed alias frequency in [0, 6] kHz and the corresponding digital frequency ω in $[0, \pi]$.

8. (a) 2 kHz
9. (b) 7 kHz
10. (c) 10 kHz
11. (d) 13 kHz
12. (e) 19 kHz

A useful procedure is: reduce modulo f_s, then fold frequencies above $f_s/2$ back toward 0.

Problem 4. Prove two analog sinusoids give the same samples [10 points]

At $f_s = 12$ kHz, compare $x_1(t)=\cos(2\pi\cdot 2000t)$ and $x_2(t)=\cos(2\pi\cdot 10000t)$.

13. (a) Write $x_1[n]$ and $x_2[n]$.
14. (b) Use a trigonometric or complex-exponential identity to prove $x_1[n]=x_2[n]$ for every integer n.
15. (c) Explain in one sentence why this proves aliasing cannot generally be undone after sampling.

Problem 5. Downsampling audio: filter first [12 points]

A recording is sampled at 48 kHz and contains useful content up to 18 kHz. You want a 16 kHz output by downsampling by 3.

16. (a) What is the new Nyquist frequency?
17. (b) Why is keeping every third sample without filtering incorrect?
18. (c) What band must be strongly attenuated before downsampling?
19. (d) If an 11 kHz sinusoid is present and you downsample naively to 16 kHz, at what frequency does it appear?

Problem 6. Ideal sinc reconstruction [12 points]

Let the discrete-time sequence be $x[n]=\delta[n]+2\delta[n-1]$, obtained from a valid bandlimited sampling process with sample period T.

20. (a) Write the ideal continuous-time reconstructed signal $x_r(t)$ using normalized $\text{sinc}(u)=\sin(\pi u)/(\pi u)$.
21. (b) Verify $x_r(0)=1$ and $x_r(T)=2$.
22. (c) Compute $x_r(T/2)$.

Problem 7. Zero-order hold frequency response [12 points]

A DAC uses a zero-order hold $h_{ZOH}(t)=u(t)-u(t-T)$.

23. (a) Derive $H_{ZOH}(j\Omega)$ by direct integration.
24. (b) Separate the result into a linear-phase term and a sinc-like magnitude term.
25. (c) Compute the normalized magnitude $|H_{ZOH}|/T$ in dB at $\omega=\Omega T=0.5\pi$ and at $\omega=\pi$.

Problem 8. Uniform quantization and SQNR [14 points]

A uniform B-bit quantizer spans approximately −1 V to +1 V. Assume the high-resolution uniform-error model when requested.

26. (a) For B=8, find the quantization step Δ.
27. (b) Find the modeled error variance $\sigma_e^2=\Delta^2/12$ and RMS quantization error.
28. (c) Estimate the SQNR of a full-scale 8-bit sine.
29. (d) Estimate the full-scale sine SQNR for 16 bits.
30. (e) Explain why “16-bit audio is about 96 dB” and “16-bit full-scale sine SQNR is about 98 dB” are not necessarily contradictory.

Problem 9. Graduate stretch — complete alias equivalence class [6 points]

At f_s = 16 kHz, a real sampled cosine is observed at an alias frequency of 3 kHz. Find every positive analog cosine frequency below 50 kHz that would produce the same sampled sequence. Derive the general frequency family rather than listing guesses.

Graduate students: treat this as required. Seniors: recommended stretch problem.

Homework — Complete Solutions

Solution 1. From continuous frequency to discrete frequency

Sampling at $f_s=16$ kHz means $T=1/16000$ s and $x[n]=x_c(nT)$.

For a continuous sinusoid at frequency f, the discrete angular frequency is $\omega=2\pi f/f_s$.

(a) $x[n] = 2\cos(2\pi\cdot 1000\, n/16000) + \cos(2\pi\cdot 5000\, n/16000) = 2\cos(\pi n/8) + \cos(5\pi n/8)$.

(b) $\omega_1 = \pi/8$ rad/sample $= 0.125\pi$; $\omega_2 = 5\pi/8$ rad/sample $= 0.625\pi$.

(c) Neither component is aliased because both analog frequencies are below $f_s/2=8$ kHz. Equivalently, both normalized frequencies lie in $[0,\pi]$.

Key answer: $x[n]=2\cos(\pi n/8)+\cos(5\pi n/8)$, with no aliasing.

Solution 2. Spectral replicas and Nyquist margin

Ideal sampling produces copies of $X_c(f)$ centered at integer multiples of f_s. Each copy has the same ±3.4 kHz support around its center.

(a) Baseband support: [−3.4, 3.4] kHz. Copies around +8 kHz occupy [4.6,11.4] kHz and around −8 kHz occupy [−11.4,−4.6] kHz. Copies around ±16 kHz occupy [12.6,19.4] kHz and [−19.4,−12.6] kHz.

(b) Adjacent replica centers are separated by $f_s=8$ kHz.

(c) The positive baseband ends at 3.4 kHz; the nearest +8 kHz replica begins at 8−3.4=4.6 kHz. The gap is 4.6−3.4=1.2 kHz.

(d) The ideal Nyquist rate is 2(3.4)=6.8 kHz. Therefore 8 kHz is above the Nyquist rate and leaves a 1.2 kHz total spectral gap between adjacent supports.

No spectral overlap occurs at 8 kHz; an ideal reconstruction filter can isolate the baseband copy.

Solution 3. Alias-frequency table

Reduce each frequency modulo 12 kHz, then use the cosine symmetry that maps any result above 6 kHz to 12 kHz minus that result. Finally use $\omega=2\pi f_{alias} / f_s$.

Analog f	Modulo 12 kHz	Alias in [0,6] kHz	ω in [0,π]
2 kHz	2	2 kHz	π/3
7 kHz	7	5 kHz	5π/6
10 kHz	10	2 kHz	π/3
13 kHz	1	1 kHz	π/6
19 kHz	7	5 kHz	5π/6

Solution 4. Prove two analog sinusoids give the same samples

At $T=1/12000$ s:

(a) $x_1[n]=\cos(2\pi\cdot 2000\, n/12000)=\cos(\pi n/3)$. Meanwhile $x_2[n]=\cos(2\pi\cdot 10000\, n/12000)=\cos(5\pi n/3)$.

(b) $5\pi n/3 = 2\pi n - \pi n/3$. Because $\cos(2\pi n-\theta)=\cos(\theta)$ for integer n, $x_2[n]=\cos(\pi n/3)=x_1[n]$. Equivalently, 10 kHz ≡ −2 kHz modulo 12 kHz and cosine is even.

(c) The same discrete sequence is consistent with two distinct analog inputs, so the samples alone cannot identify which input was present.

Aliasing is a non-uniqueness problem, not merely additive corruption.

Solution 5. Downsampling audio: filter first

After downsampling by 3, the new sample rate is 16 kHz.

(a) New Nyquist frequency = 8 kHz.

(b) The original contains energy between 8 and 18 kHz. After the sample-rate reduction, that energy would fold into [0,8] kHz unless it is removed first.

(c) A practical anti-alias filter should pass the desired content below the target passband edge and strongly attenuate content approaching and above 8 kHz. The exact transition band depends on the filter specification.

(d) 11 kHz modulo 16 kHz is 11 kHz, then folding around 8 kHz gives $16-11=5$ kHz.

Naive 48→16 kHz downsampling maps 11 kHz to a false 5 kHz component.

Solution 6. Ideal sinc reconstruction

Only two samples are nonzero, so the infinite interpolation sum collapses to two terms.

(a) $x_r(t) = \mathrm{sinc}(t/T) + 2\,\mathrm{sinc}(t/T - 1)$.

(b) At $t=0$: $\mathrm{sinc}(0)=1$ and $\mathrm{sinc}(-1)=0$, so $x_r(0)=1$. At $t=T$: $\mathrm{sinc}(1)=0$ and $\mathrm{sinc}(0)=1$, so $x_r(T)=2$.

(c) At $t=T/2$, $\mathrm{sinc}(1/2)=2/\pi$ and $\mathrm{sinc}(-1/2)=2/\pi$. Thus $x_r(T/2)=2/\pi+2(2/\pi)=6/\pi\approx 1.910$.

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Solution 7. Zero-order hold frequency response

Use the CTFT definition directly on the rectangular hold pulse.

(a) $H_{ZOH}(j\Omega)=\int_0^T e^{-j\Omega t}dt=(1-e^{-j\Omega T})/(j\Omega)$.

(b) Factor $1-e^{-j\Omega T}=e^{-j\Omega T/2}\cdot 2j\sin(\Omega T/2)$. Therefore $H_{ZOH}(j\Omega)=T\, e^{-j\Omega T/2}[\sin(\Omega T/2)/(\Omega T/2)]$.

(c) For $\omega=\Omega T=0.5\pi$: $|H|/T=\sin(\pi/4)/(\pi/4)\approx 0.9003$, or −0.912 dB. At $\omega=\pi$: $|H|/T=2/\pi\approx 0.6366$, or −3.922 dB.

ZOH droop checks: −0.91 dB at 0.5π and −3.92 dB at π.

Solution 8. Uniform quantization and SQNR

The full-scale range is 2 V, so $\Delta=2/2^B$ volts. Under the uniform-error approximation, $\sigma_e^2=\Delta^2/12$ and $e_{rms}=\Delta/\sqrt{12}$.

(a) For B=8: $\Delta=2/256=0.0078125$ V=7.8125 mV.

(b) $\sigma_e^2=(0.0078125)^2/12\approx5.086\times10^{-6}$ V^2. RMS error≈0.002255 V=2.255 mV.

(c) SQNR≈6.02(8)+1.76=49.92 dB.

(d) SQNR≈6.02(16)+1.76=98.08 dB.

(e) $20\log10(2^{16})=96.33$ dB is the binary full-scale-to-one-LSB amplitude ratio. The 98.08 dB result is the modeled power ratio between a full-scale sine (RMS=$A/\sqrt{2}$) and uniform quantization error (RMS=$\Delta/\sqrt{12}$). The reference definitions differ.

For a full-scale sine: 8-bit ≈49.9 dB SQNR; 16-bit ≈98.1 dB SQNR.

Solution 9. Graduate stretch — complete alias equivalence class

A real sampled cosine at frequency f_a is unchanged when the analog frequency is shifted by integer multiples of f_s and when its sign is reversed, because discrete-time cosine is 2π-periodic and even.

Therefore the positive analog frequencies that produce alias f_a satisfy $f = |k f_s \pm f_a|$ for integers k, retaining positive values in the requested interval.

Result With f_s=16 kHz and f_a=3 kHz, the positive frequencies below 50 kHz are 3, 13, 19, 29, 35, and 45 kHz.

General family: f = |16k ± 3| kHz. Below 50 kHz → {3, 13, 19, 29, 35, 45} kHz.

Weekly Quiz — Student Copy

Suggested time: 15 minutes. Total: 20 points. No numerical software required.

1. C/D definition [2 pts]

Write the mathematical relationship between a continuous-time signal $x_c(t)$ and its samples $x[n]$. Define T and f_s.

2. Sampled spectrum [4 pts]

For ideal impulse-train sampling, write $X_s(j\Omega)$ in terms of $X_c(j\Omega)$, T, and Ω_s. What physical picture does this equation describe?

3. Alias calculation [3 pts]

A 9 kHz sinusoid is sampled at 12 kHz. What sinusoidal frequency appears in the sampled data when interpreted in [0,6] kHz?

4. Nyquist condition [3 pts]

A signal is strictly bandlimited to 4.5 kHz. What is its ideal Nyquist rate? Is f_s=8 kHz sufficient?

5. D/A concept [2 pts]

What frequency-domain magnitude effect is introduced by a zero-order hold?

6. Quantization [3 pts]

Estimate the ideal full-scale sine SQNR for a 10-bit uniform quantizer.

7. ML bridge [3 pts]

True or false, with one-sentence justification: “Converting a 16-bit PCM recording to a float32 PyTorch tensor gives the recording 32-bit amplitude precision.”

Weekly Quiz — Solutions and Grading Guide

1. [2 pts]

$x[n]=x_c(nT)$, where T is the sampling period in seconds/sample and $f_s=1/T$ is the sampling rate in samples/s (Hz).

Grading: 1 pt relationship; 1 pt correct T/f_s definitions.

2. [4 pts]

$X_s(j\Omega)=(1/T)\,\Sigma_k\, X_c(j(\Omega-k\Omega_s))$, with $\Omega_s=2\pi/T$. It describes scaled copies of the analog spectrum shifted to every integer multiple of the sampling frequency.

Grading: 2 pts equation including $1/T$; 1 pt Ω_s; 1 pt replica interpretation.

3. [3 pts]

9 kHz at $f_s=12$ kHz folds to $|9-12|=3$ kHz.

Grading: 3 pts for 3 kHz; 2 pts if method is correct but arithmetic slips.

4. [3 pts]

Ideal Nyquist rate = $2(4.5)=9$ kHz. Therefore 8 kHz is insufficient.

Grading: 2 pts for 9 kHz; 1 pt conclusion.

5. [2 pts]

A sinc-shaped magnitude envelope (high-frequency droop), together with a linear-phase delay term.

Grading: 2 pts sinc droop; mentioning spectral images is acceptable additional detail.

6. [3 pts]

$SQNR\approx6.02(10)+1.76=61.96$ dB≈62 dB.

Grading: 1 pt formula; 2 pts numerical result.

7. [3 pts]

False. float32 changes the arithmetic/storage representation used downstream; it cannot recreate amplitude information lost when the original PCM values were quantized.

Grading: 1 pt false; 2 pts explanation that casting does not restore information.

Instructor Notes and Common Misconceptions

Nyquist terminology	Distinguish Nyquist rate (2f_M for a bandlimited signal) from Nyquist frequency (f_s/2 for a chosen sample rate).
Aliasing	Do not describe aliasing only as “distortion.” The strongest conceptual statement is non-uniqueness: distinct continuous-time inputs can generate the same samples.
Anti-aliasing order	A digital low-pass filter after an aliased ADC cannot generally separate genuine in-band energy from folded energy. The filtering has to occur before the irreversible rate change.
Sinc interpolation	The reconstruction formula is exact under ideal bandlimiting and ideal sampling. Finite sinc truncation is an approximation and can introduce ringing.
ZOH	A zero-order hold produces a continuous-time staircase. Its sinc droop is deterministic and should not be confused with quantization noise.
Quantization model	Uniform, white, signal-independent quantization error is an approximation. It is most defensible when many quantizer levels are exercised and overload is absent.
16-bit vs float32	A float32 tensor can exactly carry many 16-bit PCM codes, but casting does not increase the information content of the original measurement.
ML connection	Aliasing can create structured, repeatable spectrogram patterns that a model may learn. This makes acquisition consistency part of model validity, not merely a front-end implementation detail.